Idiom Comprehension of Chinese Patients With Aphasia: A Case Study

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Abstract

Previous studies on idiom comprehension of patients with aphasia (PWAs) mainly focused on Indo-European speakers, examining whether PWAs could correctly extract the target meaning of idioms, while among Chinese PWAs, idiom familiarity, context and other variables affecting idiom comprehension were rarely studied. Hence, this study aims to explore whether Chinese PWAs can correctly comprehend the target meaning of idioms, and further investigate the role of familiarity and context. For three Chinese PWAs, this study adopted the string-to-word matching task, taking Chinese four-character idioms as the experimental stimuli, and provided decoy words containing target meaning, literal meaning, unrelated abstract meaning and unrelated concrete meaning as the matching words of idiom items by manipulating the familiarity and contextual presence of idiom items. The results suggested that the PWAs could not correctly extract the target meaning of idioms and presented both the literal meaning tendency and the weak abstract meaning tendency, and the influence of familiarity on the comprehension of idioms was stronger than that of context. These results support the *Graded Salience Hypothesis*.

Keywords: aphasia, idiom comprehension, Chinese four-character idioms, graded salience hypothesis, case study

1. Introduction

Idioms are defined as phraseological or idiomatic units that convey speaker meaning, which cannot be determined by simply adding up the meanings of each word or morpheme (Geeraerts & Cuyckens, 2010, p. 699). Idioms are commonly used in everyday language and are typically comprehended in their figurative sense by healthy individuals without deliberate effort.

However, the comprehension of idioms can be affected by various factors, prompting

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researchers to propose several hypotheses to elucidate the underlying mechanisms of idiom comprehension. It has been observed that patients with aphasia (PWA) often struggle with comprehending idioms (Papagno & Caporali, 2007). PWA refers to a clinical patient with speech and/or comprehension disorders caused by lesions in the central nervous system and/or in cortices related to language functions. For instance, in previous studies, even when PWAs from alphabetic language countries are explicitly instructed to pay attention to the figurative meanings of idioms, they often comprehend them literally (Morawski, 2009). However, PWAs still categorize idioms with abstract lures as the same group (Lada et al., 2023). While the error inclination and the brain mechanisms underlying PWAs' idiom comprehension are topics of heated discussion, the investigation into idiom comprehension among Chinese patients, including Chinese PWAs, has been comparatively less extensive.

In the previous studies among healthy people, context and familiarity were two factors studied in idiom comprehension. Familiarity was found to have a positive correlation with the predictability of figurative expression (Morid et al., 2021; Titone & Connine, 1994). Titone and Libben (2014) found the increased familiarity of idioms facilitated priming magnitude 1000 ms post idiom offset. Context, in the realm of language comprehension and interpretation, refers to the background, prerequisites, environment, and relevant information within a given situation that help us understand the meaning and purpose of specific linguistic expressions (Lyons, 1977). Previous studies have shown that the presence of context sentences significantly improves comprehension of natural language, as individuals are better able to integrate and interpret the meaning of words within a meaningful context, leading to enhanced comprehension and reduced ambiguity (Deniz et al., 2023; Qualls et al., 2003). Holsinger (2013) discovered that contextual, lexical, and structural information provided advantages in processing idioms at early stages. Combining those two factors, familiarity was found to have stronger influence on healthy people than context (Leynes & Upadhyay, 2022), which was in line with the *Graded Salience Hypothesis*. Moreover, a recent study utilizing a self-paced reading task found that high-literality idioms were reinforced by supportive contexts, while low-literality idioms were less influenced by context (Beck & Weber, 2020). However, studies manipulating both familiarity and context to the idiom comprehension of PWAs were rarely found.

This study aims to discuss the idiom comprehension of Chinese PWAs, including their comprehension of the target meanings of idioms when there was interference and the influence of context and familiarity to their idiom comprehension. The research questions are as follows: (1) Do Chinese PWAs have deficiencies in idiom comprehension? If they do, what are these deficiencies? (2) How do context and familiarity influence the idiom comprehension of Chinese PWAs? (3) Do Chinese PWAs retain the ability to recognize the semantic category of idioms? If they do, which category do they place the idioms in?

The experiment conducted in this study was a case study. Three participants were instructed to finish a string-to-word matching task. The string part was a 2×3×2 design crossed literal cue (with literal cue vs. without literal cue), familiarity (high familiarity vs. low familiarity vs. literal paraphrase of the idioms' figurative meaning) with the context (without context vs. with context). In the word part of the task, four probe words were provided: a figurative target like "poor"

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(*pinqiong*, 贫穷), a literal cue like "live fish" (*huoyu*, 活鱼), an unrelated concrete cue like "pork" (*zhurou*, 猪肉) and an unrelated abstract cue like "important" (*zhongyao*, 重要). The participants were then asked to choose the probe word that best explained the given items. During the experiment, all participants viewed the items on a laptop screen and responded using a number pad.

Three typical Chinese PWAs were selected as participants, and Chinese four-character idioms were used as stimuli to explore the deficiencies in idiom comprehension among PWAs. The results of this case study are expected to provide valuable insights for the study of idioms among Chinese PWAs.

2. Literature Review

2.1 Idiom Comprehension

While idioms are presented in the form of phrases, previous research has revealed that they differ from other types of phrases in terms of processing and comprehension. Idioms are accessed independently, separate from the constituent words that make up the phrase (Carrol & Conklin, 2017; Chuanli et al., 2021; Conklin & Schmitt, 2008, 2012).

Various hypotheses and models have been proposed to elucidate the mechanism of idiom comprehension. Idioms were theorized to be processed similarly to clauses by employing principle-based and constraint-based approaches. The former approach prioritized syntactic computation, while the latter emphasized the contribution of linguistic information to syntactic meanings. Building upon these approaches, three models of idiom processing were put forward: the *Idiom List Hypothesis* (Bobrow & Bell, 1973) and the *Lexical Representation Hypothesis* (Swinney & Cutler, 1979) were proposed based on the former approach, and the *Literal First Hypothesis* (Moreno, 2001) was proposed based on the later approach.

The *Idiom List Hypothesis* suggests the existence of distinct literal and figurative processing strategies, arguing that idioms are stored in (and accessed from) a special list which is not part of the normal lexicon (Bobrow & Bell, 1973). The *Lexical Representation Hypothesis* holds that idioms are stored and retrieved from the lexicon in the same manner as any words (Swinney & Cutler, 1979, p. 525), refuting the notion of a special list of idioms. In this process, the initial word of an idiomatic sequence is not only retrieved for its figurative meaning but also computed for its literal meaning. However, such computation is slower than retrieval, so recognizing an idiom is faster than computing the meaning of compositional as well as literal expressions. The *Literal First Hypothesis* proposes that the processing of idioms occurs in a sequential manner, with literal meanings being processed first. Only when the result of literal processing is inconsistent with the overall meaning of the idiom, does the processing shift to activate the figurative meaning of the idiom, challenging the previous hypotheses.

Compared with the aforementioned hypotheses and models, the *Graded Salience Hypothesis* (Giora, 1997, 2004) aims to emphasize the significance of salient meanings. It posits that (1) both salient meanings and non-salient meanings contribute to the semantic processing of idioms, and (2) in the semantic processing of idioms, more salient meanings are prioritized initially. Salient meanings are habitual and regulated meanings that are commonly encountered in daily life and cultural backgrounds. In other words, salient meanings are the

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meanings that come to mind first and foremost, without the need for sequential processing. The degree of salience is associated with factors such as familiarity, conventionality, the length of context, and frequency of usage (Laurent et al., 2006). The *Graded Salience Hypothesis* has been tested by several studies. Laurent et al. (2006) recruited 30 healthy people and asked them to finish a lexical decision task: idioms of different salience were adopted as materials and participants chose the appropriate figurative or literal target to match the idioms. Experimental idioms were comparable in terms of familiarity, predictability, and latent semantic similarity between strong and weak salience. Findings showed that shorter RTs (827ms) and higher accuracy (90%) were presented by strongly salient idioms than weakly salient idioms (923ms, 59%). The behavioral results agree with the *Graded Salience Hypothesis*. Hence, familiarity, as the more salient factor in the current study, was expected to influence idiom comprehension more significantly.

2.2 Idioms in Mandarin Chinese

A Chinese idiom is a distinct linguistic unit that conveys specific meanings (Zhang et al., 2013). It typically consists of four written characters and is often derived from historical events, legendary stories, dramas, or anecdotes in novels, incorporating various cultural elements (Liu et al., 2010). Chinese idioms are characterized by their concise and succinct structure. The most prevalent form of Chinese idiom is the four-character idiom, which follows a rhythmic arrangement and possesses prosodic features with two-plus-two syllables (Zhang et al., 2013). Though there are also three-character idioms and multi-character idioms, more than 90% of Chinese idioms are composed of only four written characters, as indicated by the *Chinese Idiom Dictionary* (T. Wang, 2008; Y. Wang & Wang, 2010).

Compared to idioms in alphabetic languages, Chinese idioms have their distinct and unique characteristics, indicating the presence of specific processing models and neural mechanisms for Chinese idioms. Zhang et al. (2013) conducted a study using 108 Chinese idioms with varying degrees of compositionality, which carried identifiable parts of idioms' target meanings, as stimuli and observed graded modulations in two Event-Related Potential (ERP) components, namely N250 and N400. Based on these findings, they proposed a two-stage Chinese processing model. The first stage involves pre-lexical form representations and higher-level lexical or semantic representations, while the second stage comprises post-lexical semantic integration with the preceding context. Furthermore, several studies have investigated the unique neural mechanisms underlying the processing of Chinese idioms. In idioms of alphabetic languages, the right hemisphere has been found to play an important role. Likewise, differential patterns have been observed in the processing of Chinese idioms. Yang et al. (2016) conducted an fMRI study with 20 adult participants to examine the hemisphere mechanism of Chinese idiom processing. They found that idioms and non-idioms activated overlapping brain regions in both hemispheres. And a stronger involvement of the right brain regions, particularly areas in the parietal cortex, was observed during Chinese idiom processing in the study.

2.3 Idiom Comprehension of Patients With Aphasia

The right hemisphere is found to be important in processing figurative language (Kasparian, 2013; Sidtis & Yang, 2017). It has been suggested that the right hemisphere is

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crucial for developing a mental model of the relationship between the social environment and figurative utterances (Burgess & Chiarello, 1996). Researchers have examined idiom comprehension in patients with unilateral brain lesions and found that those with right hemispheric damage exhibit more impairments in understanding idioms than those with left hemisphere damage (Kempler et al., 1999). Furthermore, Zempleni et al. (2007) found the involvement of the right hemisphere in the processing of idioms' ambiguous information. The ambiguity of idioms was evaluated by informants to see whether they could comprehend the target meaning of idioms without a context. They found both ambiguous and unambiguous idioms were processed by both hemispheres, and that especially the right middle temporal gyrus was involved exclusively in processing the ambiguous idioms. The studies mentioned above targeted healthy people.

Additionally, studies have examined PWAs and found deficiencies in their ability to comprehend idioms (Hillert, 2004; Nenonen et al., 2002; Speer, 2013). PWAs often exhibit a tendency to interpret idioms literally. For instance, in a lexical decision task conducted by Hillert (2004), PWAs were required to determine whether a word was a pseudoword, and analysis of errors revealed a higher frequency of selections related to literal meanings. Oliver et al. (2004) conducted a study involving PWAs of various types. Participants were presented with idioms and pairs of pictures and were asked to choose the picture that corresponded to an idiom. The pair of pictures represented both the figurative and the literal meaning of the idiom. Besides the idioms, there were also several literal sentences of the same number as the idioms followed by the pair of pictures representing the target meanings and meanings that slightly differed from the target meanings. The accuracy of choosing the picture representing the literal meaning did not significantly differ between idioms and literal sentences [$F(4,56) = 1.04$, $p = .4$]. This suggests that patients with brain damage are more likely to rely on the literal choice. Moreover, this inclination towards literal interpretation in PWAs was found to be strengthened by the severity of brain lesions in the frontal and/or temporal lobes. Cacciari et al. (2006) also recruited several PWAs with mild to moderate aphasia. These participants were given a string-to-word matching task, where each idiom was embedded within a simple sentence. Words associated with the figurative meaning of the idiom string, words semantically related to the last constituent of the idiom string, and two unrelated words were provided as options. The mean number of correct responses for aphasic patients in the idiom comprehension task was 15.33 (66.65% accuracy). The researchers found a significant effect of lesion site, with patients with frontal lesions performing worse than those with non-frontal lesions [$U(6, 9) = 7.5$, $p = 0.02$].

Moreover, PWAs have been found to retain the ability to categorize idioms as abstract concepts despite their difficulties in comprehension. In a recent study, Milburn et al. (2018) used a string-to-word selection task and found that patients with aphasia demonstrated a similar accuracy rate in choosing abstract lures as they did in choosing literal lures. This suggests that idioms may be represented similarly to abstract concepts for individuals with aphasia. A previous study targeting healthy people utilizing functional magnetic resonance imaging (fMRI) suggested that the organization of categories, including objects and concepts, is predominantly situated in the temporal lobe (Bookheimer, 2002). Papagno and Caporali

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(2007) investigated the impact of transparency and ambiguity on idiom comprehension in aphasic patients. The study showed that transparency had no significant effect on idiom comprehension in these patients. However, both ambiguous and unambiguous idioms were found to be challenging for PWAs with temporal lobe lesions.

In conclusion, based on several studies conducted using Indo-European languages, it has been observed that PWAs tend to exhibit difficulties in comprehending idioms. These comprehension deficiencies seemed to be associated with the location of their lesion sites. Additionally, recent findings suggest that PWAs have a tendency to make errors in both literal and abstract interpretations of language. As mentioned before, the idiom comprehension of healthy people tends to be influenced more strongly by salient meanings of idioms than non-salient meanings. Fewer studies focused on the influence of salience to PWAs, especially Chinese PWAs. Hence, the current study is aimed to further investigate the idiom comprehension in Chinese PWAs. The study not only concentrated on whether Chinese PWAs in this study can comprehend the target meaning of idioms, but also introduced two factors which influenced the salience of idioms, familiarity and the presence of context, to study whether the impact of familiarity and context also supports the *Graded Salience Hypothesis* when participants were Chinese PWAs.

3. Methodology

3.1 Participants

This study included three right-handed participants with aphasia who were recruited from the Dalian Rehabilitation and Nursing Center. The three of them were undergoing language rehabilitation for their language deficiencies. All three were native Chinese speakers who had been diagnosed with aphasia following a cerebral stroke. The study was conducted at least six months after the onset of the aphasic disorder to ensure the stability of their condition. Their demographics and onset times are provided in Table 1. Their average age was 58.3, with a range of 36-72 years. Testing for this study took place between 6 and 21 months after the onset of aphasia. All three were able to understand the instructions for the experiments, and they did not have any significant auditory deficits or severe cognitive impairments. They were selected based on having at least 9 years of education to ensure a basic competence in Chinese. Prior to the study, the linguistic competence of the participants was evaluated using the Psycholinguistic Assessment & Treatment System in Chinese Aphasia (PACA) (J. Wang et al., 2009). The evaluation results of the linguistic modules of the participants can be found in Table 2. It's worth mentioning that S1 and S2 were unable to be evaluated in the writing module due to their paralysis. In this study, due to the limited number of cooperative hospitals and the overall shortage of PWAs and other objective reasons, only PWAs with a stable condition (months after the onset of aphasia ⩾4 months), language comprehension ability (⩾30 points for PACA reading and listening comprehension items), a high education level (⩾12 years) and who were willing to cooperate with the experiment could be selected. Hence, the factors like brain aging, gender and years of education that may influence the results would be discussed later.

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Table 1
Demographics of PWAs

Subject	Sex	Months Post-Onset	Age	Occupation	Years of Education	Lesion Site
1	M	10	67	Enterprise middle manager	12	Bilateral cortex and fronto and temporo parietal lobe
2	F	21	36	Foreign enterprise employee	16	Bilateral cortex and frontal and temporal lobes
3	F	6	72	Bank clerk	16	Left thalamus

Table 2
PACA Assessment of PWAs

Subject	Listening Comprehension	Figure Naming	Repetition	Reading	Writing
1	34/40	26/40	15/40	34/40	/
2	39/40	28/40	35/40	33/40	/
3	37/40	22/40	33/40	33/40	31/40

3.2 Stimuli

The participants were given a string-to-word matching task. During the test, each of them was presented with idioms of varying familiarity levels, as well as idioms with or without contextual information. The experiment items were adjusted based on the reading speed of each participant. The design of the experiment followed a 2 × 3 × 2 framework. This means that it involved three independent variables: literal inclination (items with a literal cue versus items without a literal cue), idiom type (high familiarity, low familiarity, and a literal paraphrase of the idioms' figurative meaning), and the presence of a figuratively-based context sentence (items without context versus items with context). To conduct the experiment, the participants were required to match the strings presented to them with the appropriate words. The strings consisted of the idioms being tested. The familiarity of the idioms, as well as the presence or absence of a literal cue and contextual information, were manipulated in order to examine their impact on the participants' performance.

The idioms used in this study were selected from the online *Chinese Idiom Dictionary* (T. Wang, 2008). These idioms consisted of four characters each like *jiatusibi,* "as poor as a house with nothing but four walls" (家徒四壁). The Chinese characters used in each idiom were chosen from frequently-used Chinese characters in the Psycholinguistic Assessment & Treatment System in Chinese Aphasia (PACA). To ensure consistency, the stroke number of each idiom was controlled and ranged from 29 to 48, with an average stroke number of 34.6. Additionally, all the idioms included at least one Chinese character that provided a literal implication of the idiom's figurative meaning. The participants then chose which of four probe words, including a figurative target like *pinqiong*, "poor" (贫穷), a literal cue like *huoyu*, "live fish" (活鱼), an unrelated concrete cue like *zhurou*, "pork" (猪肉), and an unrelated abstract cue like *zhongyao*, "important" (重要).

In this experiment, a total of 96 items were included, consisting of 72 experimental items and 24 filler items. These items were presented in two blocks, each featuring different options. Block 1 was divided into two parts, as outlined in Table 3. The first part included 36 experimental items and 12 filler items. These items were idioms of different familiarity types

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(high familiarity, low familiarity, and a literal paraphrase of the idioms' figurative meaning) presented without context. The second part of Block 1 also contained 36 experimental items and 12 filler items. The items in this part were idioms of different familiarity types presented with context. Each experimental item in both parts of Block 1 included four probe words as options: a figurative target, a literal cue, an unrelated concrete cue, and an unrelated abstract cue. Block 2 followed the same structure as Block 1 (as shown in Table 4). However, Block 2 only contained three probe words for each item: a figurative target, an unrelated concrete cue, and an unrelated abstract cue.

In the current study, the experimental stimuli consisted of four-character idioms. In Part 1 of both blocks, each experimental item was either a four-character idiom, such as "as poor as a house with nothing but four walls" (*jiatusibi*, 家徒四壁), with variable familiarity, or a word group consisting of two two-character words, such as "broke and poor" (*qiongkupinkun*, 穷苦贫困).

In Part 2 of both blocks, each experimental item consisted of a clause that included a context prime and a four-character idiom target. The context region appeared before the sentence, for example, "They had no money at all. It was simply ..." (他们家一点钱也没有，简直是……), followed by a critical region containing either the idiom itself or a literal paraphrase, such as "as poor as a house with nothing but four walls" (家徒四壁). The only variation across different conditions was in the critical idiom region.

In Block 1, the participants were instructed to choose the most appropriate probe word from an array of four options after reading each experimental item. The probe word options included a figurative target like "poor" (*pinqiong*, 贫穷), a literal cue like "fish" (*huoyu*, 活鱼), an unrelated concrete cue like "pork" (*zhurou*, 猪肉), and an unrelated abstract cue like "important" (*zhongyao*, 重要). For items that involved a literal paraphrase of the idiom, a second unrelated concrete cue like "eyeball" (*yanqiu*, 眼球) was used instead of a literal cue. This concrete cue was used to ensure that there were still four probe words for these items, allowing the participants to have the same attention resource allocation as the other items so that they were less distracted by unrelated concrete cues instead of unrelated abstract cues (Lada et al., 2023).

In Block 2, the participants were instructed to choose the most appropriate probe word from an array of three options after reading each experimental item. The probe word options in Block 2 included a figurative target like "poor" (贫穷), an unrelated concrete cue like "pork" (猪肉), and an unrelated abstract cue like "important" (重要).

In addition to the experimental items, 24 filler items were included in the study. These filler items did not contain any idioms but instead consisted of four-character words, such as "flowers of our motherland" (*zuguohuaduo*, 祖国花朵), or longer sentences like "How lively and lovely these children are, all flowers of our motherland!" (这些小朋友多么活泼、多么可爱，都是祖国花朵呀!). For the filler items, the array of probe words included a target word like "children" (*haizi*, 孩子), a literal cue selected from the item itself like "flower" (*huaduo*, 花朵), a concrete unrelated cue like "leek" (*jiucai*, 韭菜), and an abstract unrelated cue like "habitat" (*qixi*, 栖息).

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Table 3
Examples of Stimuli in Block 1

	Context	Familiarity	Idiom/Sentence With an Idiom	Figurative Target	Literal Cue	Unrelated Abstract Cue	Unrelated Concrete Cue
Part 1	Items Without Context	Low Familiarity Item	釜中生鱼	贫穷	活鱼	重要	猪肉
		High Familiarity Item	家徒四壁	贫穷	墙壁	重要	猪肉
		Literal Paraphrase Item	穷苦贫困	贫穷	眼球	重要	猪肉
Part 2	Items With Context	Low Familiarity Item	他们家一点钱也没有，简直是釜中生鱼。	贫穷	活鱼	重要	猪肉
		High Familiarity Item	他们家一点钱也没有，简直是家徒四壁。	贫穷	墙壁	重要	猪肉
		Literal Paraphrase Item	他们家一点钱也没有，简直是穷苦贫困。	贫穷	眼球	重要	猪肉

Table 4
Examples of Stimuli in Block 2

	Context	Familiarity	Idiom/Sentence With an Idiom	Figurative Target	Unrelated Abstract Cue	Unrelated Concrete Cue
Part 1	Items Without Context	Low Familiarity Item	釜中生鱼	贫穷	重要	猪肉
		High Familiarity Item	家徒四壁	贫穷	重要	猪肉
		Literal Paraphrase Item	穷苦贫困	贫穷	重要	猪肉
Part 2	Items With Context	Low Familiarity Item	他们家一点钱也没有，简直是釜中生鱼。	贫穷	重要	猪肉
		High Familiarity Item	他们家一点钱也没有，简直是家徒四壁。	贫穷	重要	猪肉
		Literal Paraphrase Item	他们家一点钱也没有，简直是穷苦贫困。	贫穷	重要	猪肉

3.3 Procedures

3.3.1 Pretests

First, a pretest, referred to as Pretest 1, was conducted using a rating questionnaire. We identified 40 pairs of synonymous idioms from the online *Chinese Idiom Dictionary* (T. Wang, 2008). The Chinese characters of each idiom were selected from frequently-used Chinese characters in Psycholinguistic Assessment & Treatment System in Chinese Aphasia (PACA), which are commonly encountered and understood in PWAs' daily rehabilitation practices. Then, a total of 30 informants were instructed to evaluate the familiarity of 40 pairs of idioms using a Likert scale ranging from 1 (very unfamiliar) to 5 (very familiar). And the top 25 pairs with the largest gaps in familiarity were selected as the experimental materials. Idioms in the lower-familiarity condition (M = 2.35; SD = 0.82) were rated as less familiar than idioms in the higher-familiarity condition (M = 4.75; SD = 0.28; $t[48] = 17.52$; $p < .001$).

Second, Pretest 2 was conducted based on the results of Pretest 1. Using the meanings provided in the online *Chinese Idiom Dictionary* (T. Wang, 2008), we generated 4 two-character words for each idiom among the top 25 pairs identified in Pretest 1. Subsequently, another 30 informants were asked to rank the 4 two-character words in terms of their similarity to the target idiom. The two-character word ranked first was used as the figurative target cue. The second and third ranked two-character words, such as *qiongku*, "broke" (穷苦) and *pinkun*, "poor" (贫困), were combined to form a four-character phrase like *qiongkupinkun*, "broke and poor" (穷苦贫困), which was used as the literal paraphrase.

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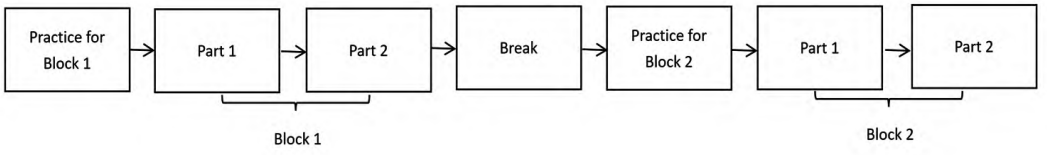
Finally, Pretest 3 was conducted, wherein each of the top 25 idiom pairs selected from Pretest 1 was placed within a context sentence devised by the researcher, following the guidelines of the Peking University CCL Corpus. The number of characters in each context sentence was controlled and ranged from 17 to 26. And another 30 informants were tasked with determining whether the context accurately matched the intended meaning of these idioms, utilizing a rating scale of 1 (match) or 0 (no match).

After conducting all the normative studies, the top 12 pairs of idioms were chosen as the experimental stimuli. The selection was based on their scores from each pretest and the need to ensure an appropriate number of Latin Square designs. The remaining pairs were utilized as practice trials in the formal experiment.

3.3.2 Formal Procedures

Prior to the experiment, the participants were instructed to view the stimuli presented on a LENOVO 21A2 laptop using Psychopy 2022 software. They were required to respond using a number pad with 4 number keys. The entire procedure is depicted in Figure 1.

Figure 1
The Procedure of the Experiment



All participants provided informed consent prior to their participation. Additionally, they were given questionnaires to collect demographic information, including age, sex, education background, and so on. Furthermore, they were instructed to complete a practice trial before the formal experiment to help them become familiar with the experimental procedures.

During the experiment, all participants were instructed to complete 2 blocks. The entire test lasted approximately 1 to 1.5 hours, including self-paced rest periods. In each block, the participants were given the following instruction:

This part will be presented as multiple-choice questions. First, an idiom will appear, followed by four two-character word options. Please choose the option that best matches the meaning of the idiom from the four options. Press 1234 to choose the answer.

The stimuli of each block were then presented sequentially, with the probe words evenly distributed spatially to prevent potential uneven cognitive distribution among the participants (Figure 2). During the experiment, all the participants viewed the items on the laptop screen and were instructed to respond using a number pad with 4 number keys, where each key had a number (1 to 4) labeled on it. However, in the case of S1 and S2 who had difficulty moving their hands due to paralysis, the experimenters assisted them by inputting their responses on the number pad based on their oral instructions.

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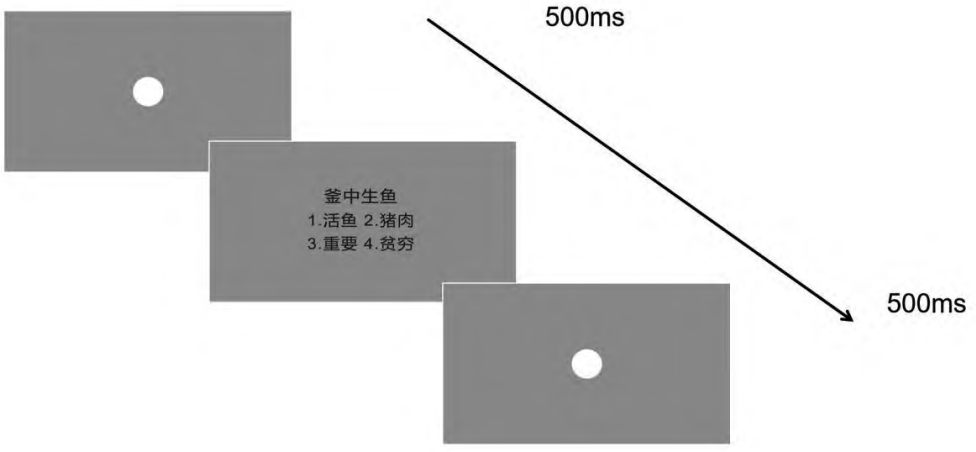
3.4 Data Analysis

The data for the normative studies were collected using Wenjuanxing, a platform in China that provides functions similar to Amazon Mechanical Turk. The data were then analyzed using the R programming language. Additionally, the data for the formal experiment were collected using Psychopy 2022. Due to the limited number of participants, the data for each of them in the formal experiment were calculated using Microsoft Excel 2022. Reaction

Figure 2
The Sequence of Stimuli Presentation



time data were excluded for participants with aphasia, as they may have limited short-term memory. Accuracy and errors were calculated and analyzed using the filtering, counting, and summation functions in the toolbar of Excel 2022.

4. Results

4.1 Literal Inclination

As shown in Table 5, when literal cues were provided, all participants exhibited lower accuracy rates compared to when there were no literal cues. Specifically, in the block with literal cues, all individuals with aphasia (PWAs) demonstrated lower accuracy compared to the block without literal cues. S1 achieved 24-out-of-72 correct responses, resulting in an accuracy rate of 33.33%. S2 showed relatively better performance, with 57-out-of-72 correct responses, corresponding to an accuracy rate of 79.17%. S3 fell in between the performance levels of the other two participants, obtaining 40-out-of-72 correct responses, yielding an accuracy rate of 55.56%.

However, when the literal cue was removed from the options, all participants demonstrated a significant improvement in accuracy rate, specifically, 53 out of 72 (73.61%), 69 out of 72 (95.83%), and 64 out of 72 (88.89%). These findings suggest that individuals with aphasia struggled to resist their inclination towards literal cues. It is important to note that

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both blocks of the study contained idioms of varying familiarity and contextual information. Therefore, we can conclude that without the intervention of removing the literal cue, individuals with aphasia found it challenging to produce accurate responses for highly familiar idioms or make correct inferences for less familiar idioms, which is in contrast to the performance of healthy individuals.

Table 5
Literal Inclination

	Subject	Correct Responses	Accuracy
With Literal Cues	1	24	33.33%
	2	57	79.17%
	3	40	55.56%
Without Literal Cues	1	53	73.61%
	2	69	95.83%
	3	64	88.89%

4.2 Abstract Inclination

With the presence of a literal cue, all PWAs demonstrated a strong inclination towards literal interpretations and showed minimal errors in abstract and concrete inclinations, as shown in Table 6. However, when the literal cue was removed, a slight difference between abstract and concrete inclinations became apparent, with S3 exhibiting a more significant abstract inclination. In the block that included a literal cue, PWAs exhibited a strong inclination towards literal interpretations and minimal errors in abstract and concrete inclinations. Nevertheless, in the block without a literal cue, a small discrepancy between abstract and concrete inclinations emerged, suggesting that PWAs showed a preference for abstract concepts only when literal interference was eliminated. It is important to note that only with human intervention could PWAs demonstrate a less pronounced inclination towards abstract concepts.

S1 consistently displayed a strong inclination towards literal interpretations as the main error tendency, with similar error tendencies in abstract and concrete inclinations. However, when there were no literal options, S1 did not show a distinct gap between abstract and concrete inclinations. S1 often muttered about the specific concrete objects he knew when choosing the concrete options. On the other hand, S2 performed the best among the three PWAs and exhibited a strong inclination towards literal interpretations without other error responses. In the block that did not contain a literal cue, S2 only selected 3 incorrect options, resulting in no significant difference between abstract and concrete inclinations. However, when the literal cues were removed, S2's inclination towards abstract interpretations became more prominent than concrete interpretations. During the experiment, S3 also demonstrated a tendency to select the concrete objects she was familiar with among the incorrect options in the concrete category.

An error analysis of literal paraphrases was conducted to highlight the distinctive impact of abstract inclination on idiom comprehension, specifically presented in Table 7. S2 exhibited flawless performance with no errors. Although S1 and S3 exhibited some errors, as indicated

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in Table 7, these errors can be considered minor and insignificant. In general, the effect of unrelated abstract and concrete concepts on PWAs' comprehension of non-figurative expressions seemed to be negligible when considering literal paraphrases. Therefore, it can be inferred that PWAs tend to rely on abstract concepts primarily when comprehending idioms.

Therefore, it can be concluded that these three participants exhibited an unrestrained inclination towards literal cues. While S3 struggled with recognizing the figurative meaning of idioms, she seemed to possess the ability to categorize idioms as abstract concepts, indicating some capacity for roughly distinguishing correct semantic categories. The few incorrect choices of concrete options might have been influenced by their tendency to recall daily necessities when presented with concrete options.

Table 6
Error Analysis

	Subject	Literal		Unrelated Abstract		Unrelated Concrete	
		Error Responses	Error Rate	Error Responses	Error Rate	Error Responses	Error Rate
With Literal Cue	1	28	58.33%	10	20.83%	10	20.83%
	2	15	100.00%	0	0.00%	0	0.00%
	3	22	78.57%	4	12.50%	2	7.14%
Without Literal Cue	1	N/A		10	52.63%	9	47.37%
	2			2	66.67%	1	33.33%
	3			7	87.50%	1	12.50%

Table 7
Error Analysis of Literal Paraphrases

Subject	Unrelated Abstract		Unrelated Concrete	
	Error Responses	Error Rate	Error Responses	Error Rate
1	4	44.44%	5	55.56%
2	0	0.00%	0	0.00%
3	1	33.33%	2	66.67%

4.3 The Influence of Familiarity on PWAs' Idiom Comprehension

Table 8 reveals an interesting observation that idioms with low familiarity exerted a similar influence to idioms with high familiarity when accuracy from both blocks was combined. This suggests that the level of familiarity did not significantly affect the accuracy of responses. Furthermore, literal paraphrases of the idioms' figurative meanings achieved the highest accuracy, indicating a notable difference only between the literal paraphrases and idioms of high familiarity. Idioms with low familiarity had a similar impact as idioms with high familiarity, while the literal paraphrases of the idioms' figurative meanings yielded the highest accuracy. Specifically, S1 and S2 were less influenced by familiarity, whereas S3 exhibited a significant difference between idioms of high familiarity and low familiarity.

The items that included literal paraphrases of the idioms' figurative meanings can be viewed as a synonym matching task, where the participants had to select the correct two-character word that had the same meaning as a four-character word group. S1 demonstrated

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an inclination towards abstract interpretations based on the errors made when choosing options consistent with the literal paraphrases of the idioms' figurative meanings. For instance, S1 selected "safe" (*anquan*, 安全) as the explanation for "diligent and industrious" (*nulikuxue*, 努力苦学). This decision was influenced by his belief that being diligent and industrious should occur in a safe environment. S1 encountered difficulties in understanding the meanings of both high- and low-familiarity idioms. The most notable errors were related to literal cues, and he also mistakenly selected many abstract cues because he excluded concrete cues, arguing that idioms were not similar to the names of everyday objects. On the other hand, both S2 and S3 performed well in selecting the correct options for the literal paraphrases of the idioms' figurative meanings, highlighting their ability to recognize synonyms. Additionally, they both achieved higher scores in idioms with high familiarity, although the significance of this finding was not as pronounced as with the literal paraphrase items.

In conclusion, the level of familiarity was found to have less influence on the comprehension of idioms among PWAs. Additionally, PWAs demonstrated fewer deficiencies in recognizing and matching synonyms, even when the synonyms were presented in different word forms. This suggests that PWAs have a relatively intact ability to understand and utilize synonyms, regardless of the specific word format or structure.

Table 8
The Accuracy in Idioms of Different Familiarity

Subject	Literal Paraphrase	Low Familiarity	High Familiarity
1	62.50%	16.67%	20.83%
2	100.00%	54.17%	83.33%
3	87.50%	25.00%	79.17%

4.4 The Influence of Context on PWAs' s Idiom Comprehension

Table 9 demonstrates that with the aid of context, individuals are able to infer the meaning of idioms even when they have low familiarity with them. However, in the case of PWAs, context appeared to have a limited contribution to their processing of idioms. When context was absent, S1 and S3 only provided 11 correct responses out of 36 (30.56%) and 19 out of 36 (52.78%) respectively. With the inclusion of context, they were able to obtain only 2 additional correct responses, indicating a 5.56% increment in accuracy that held little significance in their overall comprehension of idioms. While S3 seemed to be somewhat influenced by context, with 5 more correct responses observed in the context-present condition, the impact of context on S3's idiom comprehension remained minimal. Furthermore, S1 experienced difficulties in comprehending verbs when trying to interpret the context, which occasionally led to his reluctance in fully engaging with the experiment.

The results suggest that PWAs consistently demonstrate a slight increase in accuracy when context is provided during idiom comprehension. However, the difference between the items with context and those without context is not significant. This indicates that context contributes less to PWAs' overall comprehension of idioms. During the experiment, all PWAs read the context word by word, but it did not have a positive impact on their comprehension

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of idioms. In fact, they sometimes appeared exhausted and confused when trying to understand the context, and they also had difficulty recalling their current task or what they were doing.

Table 9
The Accuracy in Idioms With and Without Context

Subject	Without Context	With Context
1	30.56%	36.11%
2	72.22%	86.11%
3	52.78%	58.33%

5. Discussion

5.1 Deficiencies in Chinese Idiom Comprehension

The current study revealed that the three Chinese PWAs exhibited a literal inclination and were unable to comprehend idioms in a figurative manner. Similar deficiencies in idiom comprehension and literal inclination have been identified in numerous studies involving individuals with brain damage in alphabetic languages (Cacciari et al., 2006; Milburn et al., 2018; Morawski, 2009; Papagno & Caporali, 2007). The performance of Chinese PWAs in the current study aligns with the findings of these previous studies. Although idioms differ between alphabetic languages and Chinese, Chinese PWAs still exhibited deficiencies in idiom comprehension and a literal inclination.

Because the information provided of PWAs' lesion sites in the current study was limited, and because the participants were protected under military regulations, the difficulties of the participants' idiom comprehension observed in the study should be discussed about which hemisphere was damaged. Previous research has shown that individuals with right hemisphere damage tend to have more severe impairments in idiom comprehension compared to those with left hemisphere damage (Sidtis & Yang, 2017). There is also evidence suggesting the involvement of a bilateral neural network in figurative comprehension, especially the exclusive participation of the right hemisphere (Zempleni et al., 2007).

In contrast, when it comes to the unilateral damaged patients, the right hemisphere damaged patients actually performed better than the left hemisphere damaged patients (Papagno & Genoni, 2004). However, S3 with left hemisphere damage did not show significant advantages in idiom comprehension, which is different from the study targeting at unilateral damaged patients. Notably, the studies mentioned adopted idioms from alphabetic languages as stimuli. Hence, the reason for S3's performance may relate to the processing of Chinese idioms. As mentioned earlier, Chinese idioms are processed differently for bilateral involvement. Previous studies on Chinese idiom processing and comprehension have shown that both hemispheres are involved and play different roles (Yang et al., 2016). Therefore, all Chinese PWAs in this study exhibited deficiencies in idiom comprehension for they have at least one damaged hemisphere.

Among the participants in this study, S1, who had a lesion in the parietal lobe, performed the worst in idiom comprehension. This finding is consistent with previous research

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indicating that Chinese idiom comprehension requires the involvement of specific areas in the parietal cortex (Yang et al., 2016). Furthermore, in S1's case, the frontal and temporal parietal lobes were extensively injured, which could have contributed to the poor performance. Previous studies have concluded that the severity of frontal and/or temporal lesions influences idiom processing (Cacciari et al., 2006). Therefore, in the current study, PWAs with severe lesions in the fronto-temporo-parietal lobes, such as S1, were found to have greater difficulties in comprehending idioms, aligning with the findings of previous studies.

5.2 The Influence of Familiarity and Context on PWAs' Idiom Comprehension

In the current study, the factor of familiarity had a similar impact on the idiom comprehension of PWAs as it does among healthy individuals. However, while familiarity is known to be influential in the idiom comprehension of healthy individuals, PWAs were found to be less sensitive to context compared to healthy controls, supporting the *Graded Salience Hypothesis*. The influence of familiarity in healthy individuals' idiom comprehension has been supported by numerous studies. These studies have shown that high familiarity with idioms provides more advantages for comprehension compared to low familiarity (Giora, 1999; Morid et al., 2021; Titone & Libben, 2014).

Previous studies have found that patients with different brain injuries exhibit different deficiencies in processing idioms of varying degrees of familiarity. Patients with left hemisphere damage tend to perform better on high-familiar idioms, while those with right hemisphere damage perform better in processing low-familiar idioms (Vanlancker & Kempler, 1987). However, a recent study proposed that the right hemisphere is more sensitive to changes in familiarity when idioms have low ambiguity, and the right hemisphere demonstrates advantages in processing ambiguous idioms (Stephen & Sandra, 2014). In the current study, compared with S3, both S1 and S2, who had bilateral damage, exhibited less sensitivity to the variable familiarity of idioms. On the other hand, S3, with left hemisphere lesion, was more sensitive to idioms of high familiarity. As a result, S1 and S2, with bilateral damage, performed worse in comprehending idioms and were less affected by changes in familiarity, which is partly consistent with the previous study (Stephen & Sandra, 2014).

Indeed, the processing of Chinese idioms appears to have unique characteristics. A study using Chinese idioms as experimental materials found that the right upper parietal lobe and the right precuneus exhibited additional activation when participants were dealing with metaphorical language such as idioms (Yang et al., 2016). This suggests that the right brain region, particularly the parietal cortex, is extensively involved in comprehending and processing Chinese idioms. Therefore, the results of this study also provide evidence that both cerebral hemispheres should be involved in Chinese idiom comprehension, even when the idiom is highly familiar.

Context plays a significant role in idiom processing, facilitating the comprehension of idioms (Deniz et al., 2023; Holsinger, 2013; Qualls et al., 2003), especially idioms of high literalness (Beck & Weber, 2020). However, in this study, the factor of context exerted less influence on idiom comprehension regardless of the varied familiarity, agreeing with the *Graded Salient Hypothesis.* Another possible explanation for this is the characteristics of the

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Chinese language, where meaning integration does not heavily rely on the intactness of structural information. An ERP study on Chinese idioms suggested that syntactic category processing has no significant impact on semantic processing in Chinese (Liu et al., 2010). Therefore, the context in this study may have had limited advantages in comprehending idioms, regardless of whether the participants were healthy or not. Another possible reason is that most of the context in this study consisted of verbs, which can be challenging for Chinese PWAs to process. Previous studies have indicated that the word class effect in Chinese is due to the semantic differences between nouns and verbs, with verbs requiring more extensive and active brain processing, including the frontal lobe, temporal lobe, and bilateral cortex (Xia et al., 2016). Interestingly, these are also consistent with the lesion sites observed in the PWAs in the current study.

Hence, familiarity as a more salient meaning had a more positive influence on PWAs' idiom comprehension than the presence of context, supporting the *Graded Salient Hypothesis*. And the participants with bilateral damage were less sensitive to the change of familiarity than the participant with left hemisphere damage.

5.3 The Retained Ability of Semantic Category

In this study, PWAs were observed to retain the ability to classify idioms into abstract semantic categories, regardless of the context and familiarity of the idioms. This suggests that a literal inclination was not the sole error tendency when PWAs encountered idioms that were difficult for them to recognize. Instead, it appears that PWAs in this study allocated their recognition resources to roughly sorting the idioms they struggled to comprehend figuratively in the first place. This implies that while they may have had difficulties with the figurative interpretation of idioms, they still demonstrated some level of cognitive processing and categorization.

Previous research has indicated that individuals with brain damage who have aphasia generally maintain the ability to group idioms and abstract concepts together. They typically exhibit a tendency to categorize these concepts similarly (Lada et al., 2023; Papagno & Caporali, 2007). However, this study found that not all individuals with aphasia displayed a preference for abstract concepts in their error analysis. Only S3 demonstrated a tendency to group idioms and abstract concepts together when literal cues were removed. This suggests that there may be individual variations in how PWAs categorize idioms and abstract concepts.

Additionally, previous studies have found the temporal lobe was activated during the categorization of concepts (Bookheimer, 2002). In the present study, S3, who did not have a lesion in the temporal lobe, demonstrated her ability to retain semantic categories in Block 2. Conversely, S1 exhibited the poorest performance in recognizing semantic categories, likely due to the presence of a severe temporal lobe lesion. The results were in line with the findings of Bookheimer (2002).

5.4 The Influence of Other Factors on PWAs' Idiom Comprehension

In the current study, it was observed that S1 and S2 exhibited contrasting performances despite having the same lesion site. Interestingly, S2 achieved higher scores compared to S1 and S3. This discrepancy could potentially be attributed to four different factors: age-related

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brain mechanisms, literalness of experimental idioms, gender disparities, and diverse educational backgrounds.

Although S3 was the oldest participant of the three, she only had a lesion in the left hemisphere. Apart from considering the role of different hemispheres in idiom comprehension among PWAs, we also examined S1 and S2 in terms of the aging brain mechanism. The aging brain can lead to memory deterioration, attention problems, and difficulties in understanding idioms. Studies have shown that older adults tend to have more cognitive impairments (Jin et al., 2021). This decline in cognitive function can result in difficulties in suppressing irrelevant information and retrieving words from memory (Burke, 1999). Additionally, research has found that older adults tend to have difficulties producing idioms. In a study by Conner et al. (2011), a story-completion task was conducted with 40 older participants (ages 60-85) and 40 younger participants (ages 18-30), where all participants were asked to fill in the story with idioms. The results showed that the older participants had more difficulty retrieving idioms. In conclusion, even though both S1 and S2 had frontal and temporal lesions, S1 exhibited lower accuracy due to the effects of aging on the brain.

Additionally, the degree of literalness of idioms was found to influence the performance in the current study. Older individuals were found to have difficulties in processing idioms that had a literal interpretation (Westbury & Titone, 2011). In a study where younger and older adults were asked to judge whether decomposable or non-decomposable idioms had a literal meaning, older adults took longer in making their judgments. Moreover, another study found that older participants were more likely to comprehend idioms when they were presented in the context of a background story, and they took longer to comprehend idioms that were based on their literal interpretation (Haeuser et al., 2021). In the current study, idioms with high familiarity tended to have a stronger literal interpretation, and even idioms with low familiarity still had a weak literal interpretation. Therefore, S1 faced more difficulties compared to S2 due to the enhanced effects of literal interpretation in this study.

Moreover, gender was found to have an influence on idiom comprehension. Women tend to have a more global and coherent comprehension of sentences (Ferstl et al., 2008). In a study where both men and women were instructed to complete the same idiom comprehension task, it was found that women exhibited greater activation in the frontal language regions compared to men (Kana et al., 2012). Therefore, gender may be one of the factors that can explain why the male participant in the current study performed worse than the female participants.

Finally, it is significant to note that the participants in the study had varied educational backgrounds. A high level of education has been found to contribute to improved cognitive function in the aging population (Compton et al., 2000). In a study involving 102 participants ranging from 30 to 76 years old, a 60-minute battery was used to measure their memory, intelligence, and cognitive performance. The findings revealed that individuals with higher levels of education, even at an older age, showed improvements in intelligence and cognitive performance. Another study by Zhou et al. (2007) also found a positive correlation between years of education and cognitive functioning in older participants. Participants with lower levels of education exhibited a more delayed word recall ability. Therefore, in the current

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study, higher cognitive functioning likely contributed to the overall performance of S1 and S3, while S2, who had less than 4 years of education, had to put in more effort to achieve a lower accuracy.

The study has certain limitations. Firstly, the number of homogeneous participants in this study is limited. Future research should aim to recruit a larger sample size of individuals with similar characteristics to further validate the conclusions drawn from this study. Additionally, the categorization of Chinese idioms in this study did not take into account different dimensions such as ambiguity and transparency. Further investigations could categorize Chinese idioms based on these dimensions and explore their influences on PWAs' comprehension of idioms. Furthermore, it is worth noting that the PWAs in this study had a relatively higher level of education compared to participants in other studies. To gain a more comprehensive understanding, future studies could include participants with varying educational backgrounds and assess the potential impact on idiom comprehension. Lastly, while efforts were made to control the visual complexity of idioms in this study by limiting the number of strokes and using commonly used Chinese characters in PACA, further studies should aim to exercise more precise control over the number of strokes in idioms. This would ensure a more accurate examination of the effect of visual complexity on idiom comprehension. In sum, while this study sheds light on the comprehension of Chinese idioms among PWAs, its limitations, such as sample size, categorization parameters, variations in education levels, and visual complexity control, should be taken into consideration for future research endeavors.

6. Conclusions

In conclusion, this study's hypothesis yielded three significant findings. First, it was observed that Chinese PWAs in this study exhibit deficiencies in comprehending idioms. These individuals tend to rely heavily on the literal meanings of idioms and struggle to suppress this inclination when attempting to comprehend them. The severity of their idiom comprehension deficiencies can be attributed to the extent of frontal and/or temporal lesions they have sustained. Additionally, it was noted that the processing of Chinese idioms differs from that of alphabetic languages, as both hemispheres of the brain are involved in comprehension and processing. It was found that relying on the right hemisphere alone is insufficient for handling the metaphorical meanings embedded in idioms. Second, the study revealed that high familiarity has a weaker impact on idiom comprehension of the PWAs with left hemisphere damage, as opposed to contextual cues. Individuals with left hemisphere lesions were found to be more sensitive to the familiarity, while those with bilateral lesions showed insensitivity to both familiarity and context. These findings support the *Graded Salience Hypothesis*, suggesting that right hemisphere activity contributes significantly to the processing of salient meanings, such as familiarity. Finally, when the interference of literal interpretations was eliminated, Chinese PWAs in this study demonstrated the ability to classify idioms with abstract concepts into the same semantic category. This indicates a weak link between the temporal lobe and the right hemisphere of these participants, suggesting their involvement in maintaining the ability to recognize idioms' semantic categories.

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Chinese Abstracts
中文摘要

语言学视角下失语症研究可视化分析(2000–2019)

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摘要: 本文旨在从语言学的角度,分析2000年至2019年间关于失语症研究的变化与未来趋势。通过文献计量学方法,本研究回顾了3,004篇在语言学框架下包含“失语症”主题词的研究文献。研究数据来源于Web of Science数据库,涵盖了2000年至2019年间发表的相关文献。利用CiteSpace软件,本研究进行了文献共被引分析、中介中心性测量和引文突现检测,以探索和确定失语症研究领域内的主题模式、关键研究和新兴研究趋势。本研究识别出过去20年间该研究领域主要呈现出12个研究主题,研究人员重点关注失语症中的语言障碍、失语症治疗及其他相关研究议题。同时,具有高中介中心性的论文揭示了不同研究聚类之间的联系。此外,检测到的引文突现表明,旨在改善日常生活质量的治疗方法、话语任务的使用、话语治疗、在临床数据方面对原发性进行性失语症的深入探索、工作记忆评估的应用以及工作记忆训练在康复中的作用,在未来可能会持续受到更多关注。基于聚类可视化分析、中介中心性测量和引文突现检测,本文发现,语言学视角下的失语症研究更为广泛地关注沟通障碍缓解相关研究。

关键词: CiteSpace;失语症;语言学;语言障碍;失语症治疗;沟通;文献计量学综述
基金项目: 本文系辽宁省社会科学规划基金项目青年项目“老年人二语否定习得的认知神经机制及其对抑制控制能力提升研究”[项目编号:L23CYY005]的阶段性成果。

汉语失语症患者的习语理解:案例研究

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摘要: 以往对于失语症患者的习语理解研究主要以印欧语者为研究对象,关注失语症患者能否正确提取习语目标义,而对汉语失语症者以及习语熟悉度、语境等影响习语理解等变量考察不足。因此,本研究旨在探究汉语失语症患者能否正确理解习语目标义的基础上,进一步考察熟悉度与语境两种影响因素的作用。针对三位中国失语症患者,本研究采用习语词汇匹配任务,以汉语四字格成语为实验刺激,通过操纵习语项目的熟悉度和语境存在状态,提供含有目标义、字面义、无关抽象义和无关具体义的诱饵词作为习语项目的匹配词汇。结果提示失语症患者无法正确提取习语目标义且呈现字面义倾向和弱抽象义倾向,熟悉度对其习语理解影响强于语境。以上结果支持分级凸显假说。

关键词: 失语症;习语理解;汉语四字格习语;分级凸显假说;案例研究

在中国背景下对口吃者的态度:关于知识来源影响的探索性研究

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摘要: 公众的态度可能会影响口吃者的生活体验。本研究调查了中国公众对口吃者的态度。通过网络书面调查,结合封闭式和开放式问题,旨在深入了解公众态度。来自26个省和3个直辖市的102名受访者对生活中的口吃者在性格、言语、社交生活、工作、生活参与、教育、约会与婚姻、能力及沟通技巧等方面进行了评价。研究发现,公众在这些领域对口吃者普遍持有负面态度。对口吃者的态度因知识来源的不同而有所变化。与口吃者的熟悉程度可能会减少刻板印象,而来自媒体的知识可能会加剧刻板印象。文章还讨论了研究的启示及未来研究方向。

关键词: 公众态度;口吃者;性格;沟通技巧;熟悉程度;媒体

老年认知评估互动中的权势关系研究:基于多模态语用学视角

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摘要: 本研究主要考察临床语境下老年人进行认知评估时的权势关系建构问题。语料源自9名中国老年人在进行蒙特利尔认知评估(MoCA–B)(中文版)时的录音录像数据,研究通过多模态语用学视角进行注释和分析。结

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